

A Strongly Robust and Easy-Tuned Current Controller for PMSM Considering Parameters Variation

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ABSTRACT This paper proposes an improved internal-model-control (IMC)-based proportional-integral (PI) current controller for a permanent magnet synchronous motor (PMSM) drive system to solve the parameter dependence problem. The parameter sensitivity of the conventional IMC-based PI controller is analyzed, which indicates that the parameters mismatch would cause oscillation or overshoot. Therefore, this paper utilizes an uncertainty and disturbance estimator to eliminate the impact of the parameters mismatch disturbance caused by the parameters variation. The proposed controller includes an IMC-based PI controller, which is responsible for tracking, and a disturbance estimator, which deals with other control difficulties. The performance and robustness are easily achieved by separately tuning the only two parameters (the desired closed-loop bandwidth and the estimator bandwidth). The frequency-domain characteristic of the proposed current controller indicates that it can effectively suppress the parameters mismatch disturbance. Simulations and experiments have verified the effectiveness of the proposed method.

INDEX TERMS Current controller, parameters variation, PMSM, internal-model-control, frequency-domain characteristic.

I. INTRODUCTION

Thanks to the high reliability, high power factor, and high efficiency, permanent magnet synchronous motor (PMSM) has applied in important industrial applications [1]. Different types of control strategies in PMSM drive system have been proposed, including field-oriented control (FOC) [2], direct torque control (DTC) [3], [4], constant volt/frequency control (VFC), and model predictive control (MPC) [5]. FOC is the most widely used strategy among them. The stator current is decoupled into a torque current component and an excitation current component by introducing coordinate transformation. The current controller is of great importance in the control system, which should ensure the feedback current accurately follow the reference values.

Some current control strategies have played an essential role in the PMSM drive system, such as proportional-integral

(PI) control, hysteresis control [6], and predictive control [7]. The PI control is widely applied under synchronous rotation coordinates thanks to many advantages, including fixed switching frequency, strong robustness, and zero steady-state error. However, the PMSM control system is a nonlinear system with unavoidable disturbances, as well as parameters variation. It is a challenge to determine the PI controller parameters for obtaining a satisfying dynamic performance in the full range of operations. IMC provides a trade-off between closed-up performance and robustness with a single tuning parameter [8]. Therefore, L. Harnefors and H.-P. Nee proposed an internal model control (IMC)-based PI current controller [9]. The IMC-based PI current controller represents the proportional gain and integral gain of PI controller as an algebraic expression of both machine parameters and desired closed-loop bandwidth. Based on this conclusion, the parameters tuning is regarded as a single parameter (expected closed-loop bandwidth) optimization problem if the machine parameters are accurately known. However, this method depends on specific machine parameters to achieve

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the pole-zero cancellation. Otherwise, the output would be oscillation or overshoot [10].

Several solutions are emerging to suppress the mismatch disturbance caused by parameters variation. These methods can be classified into three groups. The first method tunes the PI gains online though some techniques such as parameters estimation method [11] and neuro-fuzzy technique [12]. However, this kind of method is quite time-consuming, which results in a low utilizing rate of hardware resource when the operating conditions change slowly. The second method replaces PI controller with advanced control strategies such as model-free predictive control [13] and adaptive sliding mode control [14]. These methods have improved the control performance in some aspects, faster transient-state response, or better robustness against machine parameter variations, but the superiority comes at the expense of higher current controller complexity. The third method comes from disturbance observers such as disturbance observer [15], [16], Luenberger observer [17] and uncertainty and disturbance estimator (UDE) [18]–[20]. These methods consider the effect of parameters mismatch and uncertainties as lumped disturbance, and then the disturbance observer is applied to estimate the adverse effect. The disturbance observers are combined with different control strategies such as model predictive control, which requires additional resources to calculate the control signals. This will bring more uncertainties into the system, which is unacceptable.

The PID controller is still dominant in industrial applications for its simplicity and ease of use [21]. This provides strong incentives to explore current regulation schemes with similar structural simplicity and dynamic response. Therefore, this paper attempts to accommodate the deterioration of the IMC-based PI current controller with the assistance of UDE.

The feature that makes the proposed method distinct from other advanced methods is attributed to the engineering friendliness. The proposed strategy needs a negligible amount of computation and is entirely compatible with existing control systems. Different from the UDE-based method in [20], the control law used in this paper makes the UDE can be embedded into the IMC-based PI current controller in a bumpless manner without retuning the PI parameters. The IMC-based PI controller is responsible for tracking, and the estimator is in charge of the other control difficulties. The performance and robustness are achieved easily by tuning the only two parameters (the desired closed-loop bandwidth and the estimator bandwidth), separately.

This paper is organized as follows. Section II introduces the traditional IMC-based PI controller and analyzes the influences of machine parameters variation on system. Section III proposes the innovative current controller and investigates its robustness. Section IV gives a parameters tuning method. Section V verifies the effectiveness of the proposed controller by comparative experiments. Finally, section VI concludes this paper.

II. IMC-BASED PI CONTROLLER

A. DESIGN OF THE CONTROLLER

The voltage equations of stator winding of a PMSM in a dq -axes frame are given by

$$\begin{cases} u_d(t) = R_s i_d(t) + L_d (di_d(t)/dt) - L_q \omega_r i_q(t) \\ u_q(t) = R_s i_q(t) + L_q (di_q(t)/dt) + L_d \omega_r i_d(t) + \psi_f \omega_r \end{cases} \quad (1)$$

where u_d , u_q , i_d , i_q , L_d , and L_q are the voltage, current, and inductance of stator winding in the dq -axes frame, respectively; R_s is the resistance of stator winding; ψ_f is the permanent magnet's flux linkage; ω_r is the electrical angular velocity.

Decoupling the machine dynamics by letting

$$\begin{bmatrix} u_d(t) \\ u_q(t) \end{bmatrix} = \begin{bmatrix} v_d(t) \\ v_q(t) \end{bmatrix} + \begin{bmatrix} -L_q \omega_r i_q(t) \\ L_d \omega_r i_d(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \psi_f \omega_r \end{bmatrix} \quad (2)$$

where $v_d(t)$, $v_q(t)$ are the outputs of the dq -axes controller, respectively. This decoupling is regarded as an internal feedback loop.

Substituting (2) into (1) results in

$$\begin{bmatrix} v_d(t) \\ v_q(t) \end{bmatrix} = R_s \begin{bmatrix} i_d(t) \\ i_q(t) \end{bmatrix} + \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} \begin{bmatrix} di_d(t)/dt \\ di_q(t)/dt \end{bmatrix} \quad (3)$$

Taking the Laplace transform on its both sides, then

$$\begin{bmatrix} I_d(s) \\ I_q(s) \end{bmatrix} = \mathbf{G}(s) \begin{bmatrix} V_d(s) \\ V_q(s) \end{bmatrix} \quad (4)$$

$$\text{where } \mathbf{G}(s) = \begin{bmatrix} 1/(R_s + sL_d) & 0 \\ 0 & 1/(R_s + sL_q) \end{bmatrix}.$$

According to pole-zero cancellation, design the PI controller as follows

$$\mathbf{C}(s) = \begin{bmatrix} K_d(1 + 1/sT_{id}) & 0 \\ 0 & K_q(1 + 1/sT_{iq}) \end{bmatrix} \quad (5)$$

where $K_d = \alpha L_d$, $K_q = \alpha L_q$, $T_{id} = L_d/R_s$, $T_{iq} = L_q/R_s$, α is the desired closed-loop bandwidth of current loop.

The open-loop transfer function of current loop at unit negative feedback is

$$\mathbf{G}_o(s) = \mathbf{C}(s) \mathbf{G}(s) = \begin{bmatrix} \alpha/s & 0 \\ 0 & \alpha/s \end{bmatrix} \quad (6)$$

Further, the closed-loop transfer function of the current loop at unit negative feedback is

$$\mathbf{G}_{cl}(s) = \frac{\mathbf{G}_o(s)}{1 + \mathbf{G}_o(s)} = \begin{bmatrix} \alpha/(s + \alpha) & 0 \\ 0 & \alpha/(s + \alpha) \end{bmatrix} \quad (7)$$

The closed-loop transfer function of current loop is a first-order element with the bandwidth α . However, zeros and poles cancellation does not complete since the inductance and resistance of stator winding change in a complex operating environment. The next part will analyze the parameter sensitivity of the IMC-based PI current controller.

B. PARAMETER SENSITIVITY ANALYSIS

According to (5), the IMC-based PI controller depends on PMSM model parameters including stator resistance and stator inductance. In a practical control system, for example, the value of stator inductance decreases as the stator current rises, and the value of stator resistance increases as the stator temperature rises. In addition, these crude parameter values are initially inaccurate owing to an offline measurement error. In this case, an analysis of the IMC-based PI controller parameter sensitivity is necessary.

Considering the machine parameters vary during the regular operation, the matrix $\mathbf{G}(s)$ in (4) may become

$$\mathbf{G}'(s) = \begin{bmatrix} 1/R'_s + sL'_d & 0 \\ 0 & 1/R'_s + sL'_q \end{bmatrix} \quad (8)$$

where R'_s , L'_d and L'_q are the machine parameters after the changes.

However, the PI controller uses the original machine parameters R_s , L_d and L_q . Therefore, the PI controller $\mathbf{C}'(s)$ is the same as the original.

$$\mathbf{C}'(s) = \begin{bmatrix} \alpha(L_d + R_s/s) & 0 \\ 0 & \alpha(L_q + R_s/s) \end{bmatrix} = \mathbf{C}(s) \quad (9)$$

Considering the parameters variation, the closed-loop transfer function of the current loop at unit negative feedback is

$$\mathbf{G}'_{cl}(s) = \frac{\mathbf{C}'(s) \mathbf{G}'(s)}{1 + \mathbf{C}'(s) \mathbf{G}'(s)} = \begin{bmatrix} G'_d(s) & 0 \\ 0 & G'_q(s) \end{bmatrix} \quad (10)$$

where

$$G'_d(s) = \frac{\alpha L_d s + \alpha R_s}{L'_d s^2 + (R'_s + \alpha L_d) s + \alpha R_s}$$

$$G'_q(s) = \frac{\alpha L_q s + \alpha R_s}{L'_q s^2 + (R'_s + \alpha L_q) s + \alpha R_s}$$

Given that $L'_d = 2L_d$, $L'_q = 0.5L_d$, $R'_s = 2R_s$, and $R'_s = 0.5R_s$, respectively. Fig. 1 shows the Bode diagram

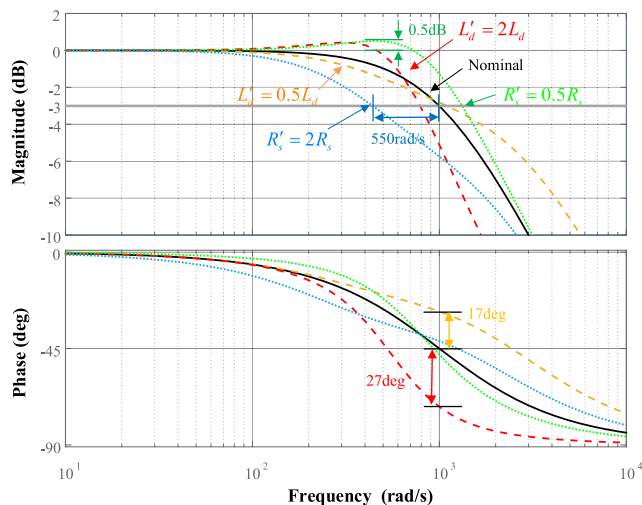


FIGURE 1. Bode diagram of the transfer function $G'_d(s)$.

of the transfer function $G'_d(s)$. And the Bode diagram of the transfer function $G'_q(s)$ is the same as $G'_d(s)$.

The parameters variation causes the closed-loop transfer function to increase by a zero and a pole, which results in an imperfect system response. Fig. 1 indicates that when the resistance reduces by half, the amplitude-frequency characteristic curve of current loop transfer function has an overshoot of 0.5 dB, which causes the current response to oscillate or overshoot. When the resistance doubles, the current loop bandwidth drops by 550 rad/s, which makes the current response slow and tracking performance terribly. The inductance variation mainly affects the phase of the current loop. The phase lag of the original system alleviates with the decrease of the inductance. On the contrary, the phase lag aggravates.

IMC-based PI controller has weak suppression ability to the mismatch disturbance according to the above analysis. Therefore, this paper uses uncertainty and disturbance estimator to estimate the disturbance, which makes up for the defects of the IMC-based PI controller.

III. INNOVATIVE CURRENT CONTROLLER

A. THE STRUCTURE OF THE CONTROLLER

Given that the machine parameters have an increment based on the original values, (1) is redefined by

$$\begin{cases} u_d(t) = (R_s + \Delta R) i_d(t) + (L_d + \Delta L_d) (di_d(t)/dt) \\ \quad - (L_q + \Delta L_q) \omega_r i_q(t) \\ u_q(t) = (R_s + \Delta R) i_q(t) + (L_q + \Delta L_q) (di_q(t)/dt) \\ \quad + (L_d + \Delta L_d) \omega_r i_d(t) + \psi_f \omega_r \end{cases} \quad (11)$$

where ΔR , ΔL_d , and ΔL_q are the increment of the stator winding resistance, d -axis inductance, and q -axis inductance, respectively. $\Delta R = R'_s - R_s$, $\Delta L_d = L'_d - L_d$, and $\Delta L_q = L'_q - L_q$.

Decoupling (11) by (2), then

$$\begin{bmatrix} di_d(t)/dt \\ di_q(t)/dt \end{bmatrix} = \mathbf{A} \begin{bmatrix} i_d(t) \\ i_q(t) \end{bmatrix} + \mathbf{B} \begin{bmatrix} v_d(t) \\ v_q(t) \end{bmatrix} + \mathbf{f}(t) \quad (12)$$

where $\mathbf{A} = \begin{bmatrix} -R_s/L_d & 0 \\ 0 & -R_s/L_q \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 1/L_d & 0 \\ 0 & 1/L_q \end{bmatrix}$

The uncertainties caused by the parameters mismatch $\mathbf{f}(t)$ is

$$\mathbf{f}(t) = \begin{bmatrix} -\Delta R/L_d & \Delta L_q \omega_r \\ -\Delta L_d \omega_r & -\Delta R/L_q \end{bmatrix} \begin{bmatrix} i_d(t) \\ i_q(t) \end{bmatrix} + \begin{bmatrix} -\Delta L_d/L_d & 0 \\ 0 & -\Delta L_q/L_q \end{bmatrix} \begin{bmatrix} di_d(t)/dt \\ di_q(t)/dt \end{bmatrix} \quad (13)$$

Let $\mathbf{x} = [i_d \ i_q]^T$, and $\mathbf{u} = [v_d \ v_q]^T$, (12) can be written as follows

$$\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t) + \mathbf{B} \mathbf{u}(t) + \mathbf{f}(t) \quad (14)$$

Then $\mathbf{f}(t)$ is given by

$$\mathbf{f}(t) = \dot{\mathbf{x}}(t) - \mathbf{A} \mathbf{x}(t) - \mathbf{B} \mathbf{u}(t) \quad (15)$$

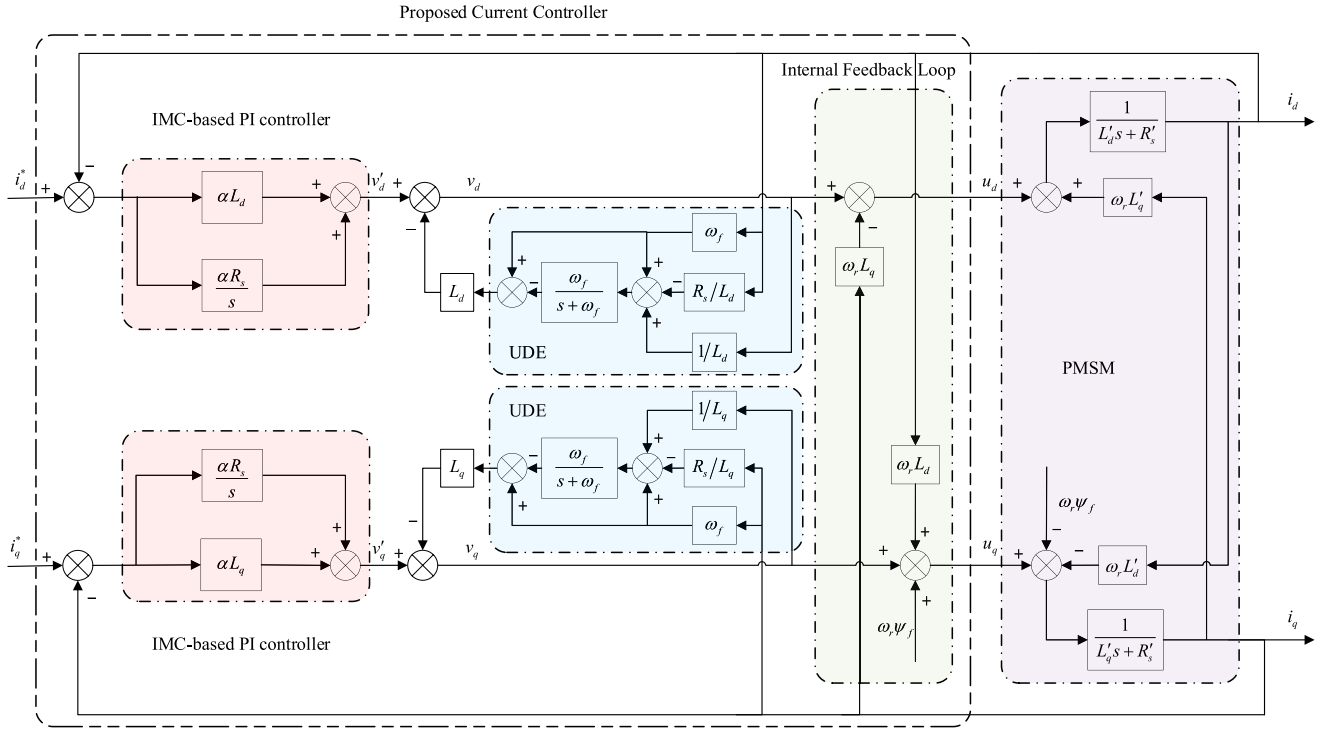


FIGURE 2. Block diagram of the proposed current control system.

However, it is difficult to deal with (15) since $\dot{\mathbf{x}}(t)$ cannot be measured directly. Follow the design procedures presented in [18], if a filter $g_f(t)$ has the unity gain and zero phase shift over the full frequency range of $\mathbf{f}(t)$, then the estimated value $\hat{\mathbf{f}}(t)$ of $\mathbf{f}(t)$ can be approximated as

$$\hat{\mathbf{f}}(t) = \mathbf{f}(t) * g_f(t) \quad (16)$$

where “ $*$ ” is the convolution operator.

Here, a first-order low-pass filter $G_f(s)$ is chosen

$$G_f(s) = \omega_f / (s + \omega_f) \quad (17)$$

where $G_f(s)$ is the Laplace transform of $g_f(t)$, ω_f is the bandwidth of $G_f(s)$. Although it is challenging to design a strictly suitable filter, the controller can compensate for the error caused by the filter.

Design the feedforward path to offset the uncertainties after estimating them, let

$$\mathbf{u}(t) = \mathbf{u}'(t) - \hat{\mathbf{f}}(t) / \mathbf{B} \quad (18)$$

where $\mathbf{u}'(t)$ is the output of the PI controller.

Substituting (18) into (14) results in

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}[\mathbf{u}'(t) - \hat{\mathbf{f}}(t) / \mathbf{B}] + \mathbf{f}(t) \\ &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}'(t) + [\mathbf{f}(t) - \hat{\mathbf{f}}(t)] \end{aligned} \quad (19)$$

If the estimation error is small enough, (19) can be transformed as

$$\dot{\mathbf{x}}(t) \approx \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}'(t) \quad (20)$$

Since (20) is similar with (3), the PI controller can be designed by referring to (8), and then its output is

$$\begin{aligned} \mathbf{U}'(s) &= (\mathbf{X}^*(s) - \mathbf{X}(s)) \mathbf{C}'(s) \\ &= (\mathbf{X}^*(s) - \mathbf{X}(s)) (\mathbf{B}^{-1} \alpha (1 - \mathbf{A}/s)) \end{aligned} \quad (21)$$

where $\mathbf{U}'(s)$, $\mathbf{X}^*(s)$, and $\mathbf{X}(s)$ are the Laplace transform of $\mathbf{u}'(t)$, given current, and feedback current, respectively.

Substituting (15), (16) into (18) and taking the Laplace transform on its both sides, then

$$\mathbf{U}(s) = \mathbf{U}'(s) - \mathbf{B}^{-1}(s\mathbf{X}(s) - \mathbf{A}\mathbf{X}(s) - \mathbf{B}\mathbf{U}(s)) G_f(s) \quad (22)$$

where $\mathbf{U}(s)$ is the Laplace transform of $\mathbf{u}(t)$.

Rewriting (2) in the s -domain form

$$\mathbf{U}_0(s) = \mathbf{U}(s) + \mathbf{D}(s) \quad (23)$$

where $\mathbf{U}_0 = [u_d \ u_q]^T$, $\mathbf{D}(s) = [-L_q \omega_r i_q \ L_d \omega_r i_d + \psi_f \omega_r]^T$

Considering (17), (21), and (22), the control law (23) can be rearranged as (24), as shown at the bottom of the next page.

Fig. 2 describes the overall block diagram of the proposed current control system. The proposed current controller consists of three parts:

- 1) IMC-based PI controller is responsible for zero-offset reference tracking and designing the closed-loop transfer function of the system as a first-order element.
- 2) UDE is expected to estimate the uncertainties caused by parameters variation, including the effect on the coupling terms.
- 3) Internal feedback loop decouples dq -axes of the motor.

B. CURRENT LOOP TRANSFER FUNCTION

Substituting (21) into (19) results in

$$\mathbf{X}(s) = \mathbf{D}_x \mathbf{X}^*(s) + \mathbf{D}_f \mathbf{F}(s) \quad (25)$$

where

$$\mathbf{D}_x = \frac{\mathbf{B}\mathbf{C}'(s)}{s\mathbf{I} - \mathbf{A} + \mathbf{B}\mathbf{C}'(s)} = \frac{\alpha}{s + \alpha} \mathbf{I}_{2 \times 2}$$

$$\mathbf{D}_f = \frac{1 - G_f(s)}{s\mathbf{I} - \mathbf{A} + \mathbf{B}\mathbf{C}'(s)} = \mathbf{L}\mathbf{H}_1\mathbf{H}_2$$

where $\mathbf{I}_{2 \times 2}$ is unit diagonal matrix,

$$\mathbf{L} = \frac{1}{s + \alpha} \mathbf{I}_{2 \times 2}, \quad \mathbf{H}_1 = \frac{s}{s + \omega_f} \mathbf{I}_{2 \times 2}$$

$$\mathbf{H}_2 = \begin{bmatrix} s/(s + R_s/L_d) & 0 \\ 0 & s/(s + R_s/L_q) \end{bmatrix}$$

Equation (25) indicates that the transfer function matrix $\mathbf{D}_x$ from given current $\mathbf{X}^*(s)$ to output current $\mathbf{X}(s)$ is consistent with $\mathbf{G}_{cl}(s)$. For a well-designed current control system, the ideal closed-loop transfer function is also a first-order element.

The transfer function matrix $\mathbf{D}_f$ from uncertainties $\mathbf{F}(s)$ to output current $\mathbf{X}(s)$ consists of two high-pass filters $\mathbf{H}_1$ and $\mathbf{H}_2$ and a low-pass filter $\mathbf{L}$. The high-frequency components and low-frequency components of the uncertainty are attenuated by the low-pass filter and the two high-pass filters, respectively. Fig. 3 shows the amplitude-frequency curves of $\mathbf{L}$, $\mathbf{H}_1$, $\mathbf{H}_2$ and $\mathbf{D}_f$. The amplitude of the low-pass filter $\mathbf{L}$ in the low frequency decreases as α increases, and the system's ability to resist the disturbance is enhanced. The magnitude of $\mathbf{D}_f$ decreases as ω_f increases, especially in the low frequency range. In general, the larger the ω_f is, the better the controller attenuates the uncertainties, and the smaller the influence of the machine parameters variation on the system.

Substituting the Laplace transform of (13) into (25), the current loop transfer function with parameters mismatch is given by

$$\mathbf{G}_{cl}''(s) = \begin{bmatrix} G_d''(s) & 0 \\ 0 & G_q''(s) \end{bmatrix} \quad (26)$$

where $G_d''(s)$ and $G_q''(s)$, as shown at the bottom of this page.

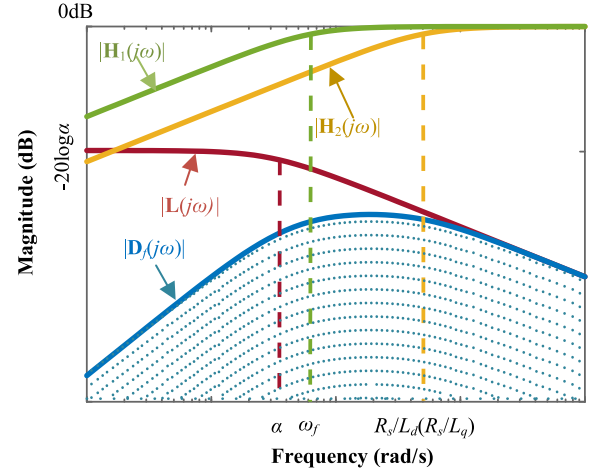


FIGURE 3. The amplitude-frequency characteristic curves of $\mathbf{L}$, $\mathbf{H}_1$, $\mathbf{H}_2$ and $\mathbf{D}_f$.

Given that $L_d' = 2L_d$, $L_d'' = 0.5L_d$, $R_s' = 2R_s$, and $R_s'' = 0.5R_s$, respectively. Fig. 4 shows the Bode diagram of the transfer function $G_d''(s)$. ($G_q''(s)$ is the same as $G_d''(s)$).

Table 1 illustrates the effect of parameters variation on closed-loop transfer function using conventional controller (Fig. 1) and the innovative controller (Fig. 4). “Overshoot” refers to the maximum value of magnitude exceeding 0dB. “Bandwidth drop” refers to the cutoff frequency of the original system (1000rad/s) minus the cutoff frequency of the system with parameters variation. “Phase change” refers to the difference of phase between the two system at 1000 rad/s. The absolute values of the three indicators represent the impact of parameters mismatch on the system. The absolute values of the proposed method are smaller than the conventional controller. Furthermore, they get smaller as the parameter ω_f increases. For example, the “bandwidth drop” of the conventional method with resistance mismatch ($R_s' = 2R_s$) is 550rad/s, while it drops to 143rad/s ($\omega_f = 2000$ rad/s) and 70rad/s ($\omega_f = 4000$ rad/s) after adopting the proposed method. Therefore, the proposed current controller has better adaptability to machine parameters variation. Additionally, the larger the ω_f , the stronger the robustness.

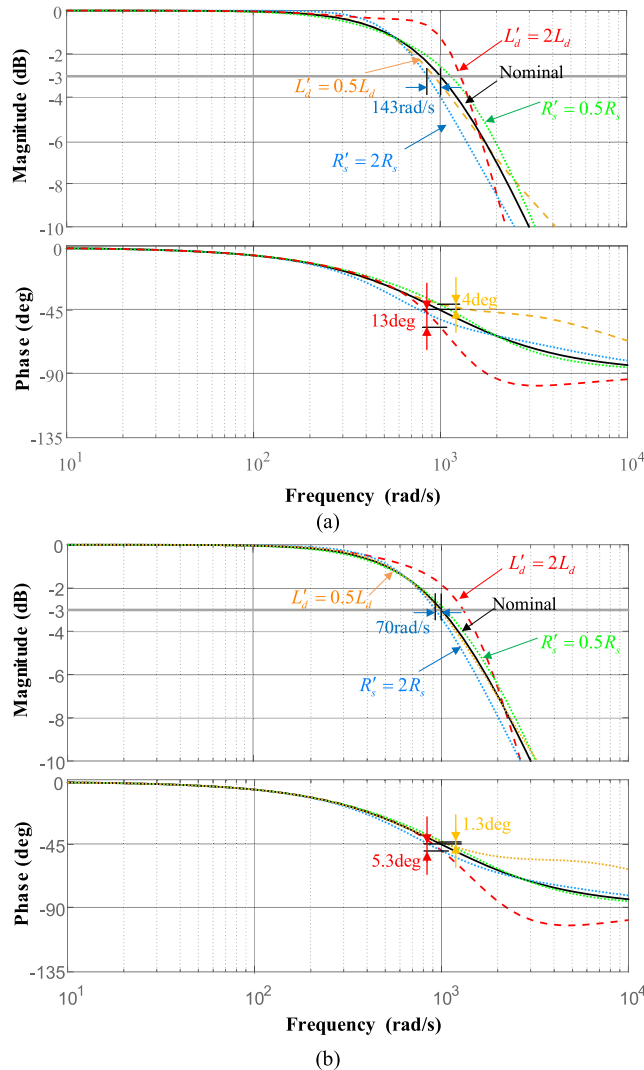
$$\begin{aligned} \mathbf{U}_0(s) &= \mathbf{U}(s) + \mathbf{D}(s) \\ &= (\mathbf{X}^*(s) - \mathbf{X}(s)) \left(\mathbf{B}^{-1} \alpha (1 - \mathbf{A}/s) \right) - \mathbf{B}^{-1} (s\mathbf{X}(s) - \mathbf{A}\mathbf{X}(s) - \mathbf{B}\mathbf{U}(s)) \cdot \omega_f / (s + \omega_f) + \mathbf{D}(s) \\ &= \underbrace{(\mathbf{X}^*(s) - \mathbf{X}(s)) \left(\mathbf{B}^{-1} \alpha (1 - \mathbf{A}/s) \right)}_{\text{IMC-based PI}} - \underbrace{\mathbf{B}^{-1} [\omega_f \mathbf{X}(s) - (\omega_f \mathbf{X}(s) + \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)) \cdot \omega_f / (s + \omega_f)]}_{\text{UDE}} + \underbrace{\mathbf{D}(s)}_{\text{Internal feedback loop}} \end{aligned} \quad (24)$$

$$G_d''(s) = \frac{\alpha (L_d s^2 + (L_d \omega_f + R_s) s + \omega_f R_s)}{L_d' s^3 + (\alpha L_d + \omega_f L_d + R_s') s^2 + (\alpha \omega_f L_d + \alpha R_s + \omega_f R_s) s + \alpha \omega_f R_s}$$

$$G_q''(s) = \frac{\alpha (L_q s^2 + (L_q \omega_f + R_s) s + \omega_f R_s)}{L_q' s^3 + (\alpha L_q + \omega_f L_q + R_s') s^2 + (\alpha \omega_f L_q + \alpha R_s + \omega_f R_s) s + \alpha \omega_f R_s}$$

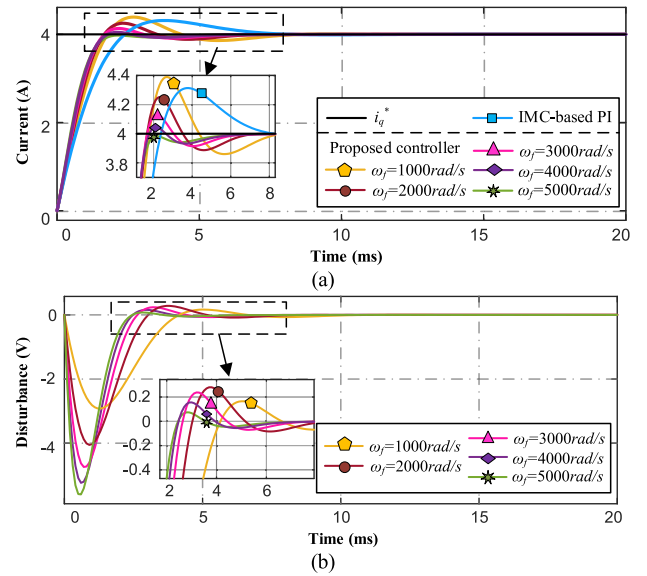
TABLE 1. The effect of parameter variation on closed-loop transfer function using conventional controller and the innovative controller.

Controller		$L'_d = 2 L_d$			$L'_d = 0.5 L_d$		
		Overshoot (dB)	Bandwidth drop (rad/s)	Phase change (deg)	Overshoot (dB)	Bandwidth drop (rad/s)	Phase change (deg)
IMC-based PI controller		0.4	200	27	0	-50	-17
Proposed current controller	$\omega_f = 2000$ rad/s	0	-35	13	0	70	-4
	$\omega_f = 4000$ rad/s	0	-35	5.3	0	20	-1.3
Controller		$R'_s = 2 R_s$			$R'_s = 0.5 R_s$		
		Overshoot (dB)	Bandwidth drop (rad/s)	Phase change (deg)	Overshoot (dB)	Bandwidth drop (rad/s)	Phase change (deg)
IMC-based PI controller		0	550	-3.4	0.5	-40	2.5
Proposed current controller	$\omega_f = 2000$ rad/s	0	143	4	0	-21	-4
	$\omega_f = 4000$ rad/s	0	70	3.7	0	-4	-1.1

**FIGURE 4.** Bode diagram of the transfer function $G_d''(s)$. (a) $\omega_f = 2000$ rad/s (b) $\omega_f = 4000$ rad/s.

IV. PARAMETERS TUNING

The proposed current controller contains two parameters: α and ω_f . α is the closed-loop bandwidth of the current loop, and ω_f is the bandwidth of the first-order low-pass filter

**FIGURE 5.** Simulating results with different ω_f under inductance mismatch ($L_s = 0.5 L'_s$). (a) current step response (b) estimated disturbance.

$G_f(s)$. The MATLAB/Simulink is used to analyze the effect of the controller parameters on the current response. The used motor is a surface-mount PMSM, $L_d = L_q$.

The effect of ω_f on the transient response with mismatch is tested firstly. As depicted in Fig. 5, keeping $\alpha = 1500$ rad/s, the overshoot and settling time get smaller as ω_f increases with inductance mismatch ($L_s = 0.5 L'_s$). Because the surface-mount PMSM is chosen as a control target, $L_s = 0.5 L'_s$ means that $L_d = 0.5 L'_d$, $L_q = 0.5 L'_q$. Table 2 lists the overshoot and settling time.

As shown in (13), the q -axis disturbance caused by the inductance mismatch is related to the differential of i_q when $i_d = 0$. The resistance mismatch disturbance is a constant value. The larger ω_f is helpful to estimate the disturbance more quickly, as shown in Fig. 5(b).

The same conclusion is obtained when considering both resistance and inductance variations, as shown in Fig. 6 and TABLE 3.

TABLE 2. Current response performance index with inductance mismatch $L_s = 0.5 L'_s$.

Performance Index	IMC-based PI	Proposed controller with different ω_f (rad/s)				
		1000	2000	3000	4000	5000
Overshoot (A)	7.8%	10.0%	6.25%	3.25%	1.0%	0%
Settling time (ms)	7.1	7.6	6	5.2	1.7	1.9

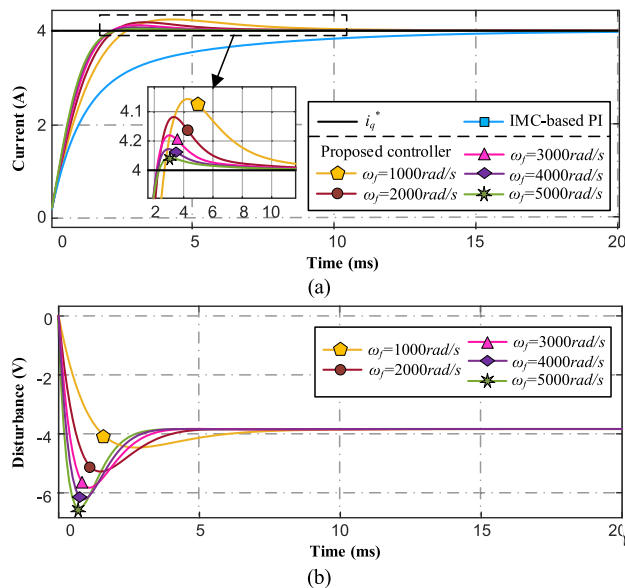


FIGURE 6. Simulating results with different ω_f under mismatch ($L_s = 0.5L'_s$, $R_s = 0.3R'_s$) (a) current step response (b) estimated disturbance.

TABLE 3. Current response performance index with both inductance mismatch ($L_s = 0.5 L'_s$) and resistance mismatch ($R_s = 0.3 R'_s$).

Performance Index	IMC-based PI	Proposed controller with different ω_f (rad/s)				
		1000	2000	3000	4000	5000
Overshoot (A)	0	6%	4.5%	3%	1.75%	1%
Settling time (ms)	17	9.6	6.5	5.1	4.3	2.1

Thus, $\omega_f = 4000$ rad/s is selected for proper overshoot and settling time while both the resistance and inductance mismatch exist and $\alpha = 1500$ rad/s.

With the same method, the effect of α is achieved. Fig. 7 denotes that the larger α , the shorter the settling time. However, there exist more overshoot as the parameter α increases on account of the larger differential of i_q . Therefore, a larger ω_f should be selected to maintain the current tracking performance when α increases.

Based on the above analysis, the parameters ($\alpha = 1500$ rad/s, $\omega_f = 4000$ rad/s) of the proposed controller are determined. Compared with the traditional IMC-based

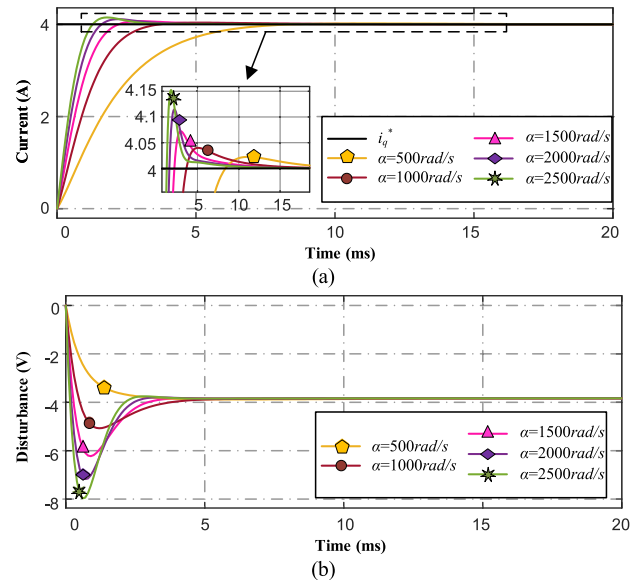


FIGURE 7. Simulating results with different α under mismatch ($L_s = 0.5L'_s$, $R_s = 0.3R'_s$) (a) current step response (b) estimated disturbance.

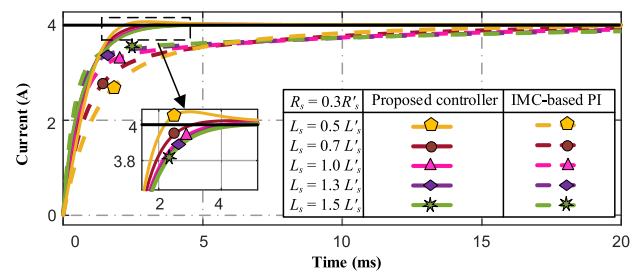


FIGURE 8. Simulating results of the proposed method with $\alpha = 1500$ rad/s and $\omega_f = 4000$ rad/s under different mismatch.

PI controller ($\alpha = 1500$ rad/s), the proposed method can obtain better transient response under resistance mismatch and different inductance mismatches, as shown in Fig. 8. The better transient response and zero-static-error characteristic can be achieved on account of the proposed method when the inductance ranges from $0.5L'_s$ to $1.5L'_s$.

In summary, the tuning procedures are as follows

- 1) Select the initial values, $\alpha = \alpha_0$, $\omega_f = \omega_{f0}$.
- 2) Increase ω_f until the change of settling time can be ignored.
- 3) Increase α if the tracking performance does not meet the requirement.
- 4) Decrease α and ω_f if there is too much noise in u_d , u_q .
- 5) Repeat the above steps.

V. EXPERIMENT AND ANALYSIS

An experimental platform with an F28069M control chip and DRV8312 driver chip is selected to validate the proposed current controller, as shown in Fig. 9. The machine parameters are in Table 4. The system sampling period and dc-bus voltage are set to $100\mu s$ and 24V, respectively.

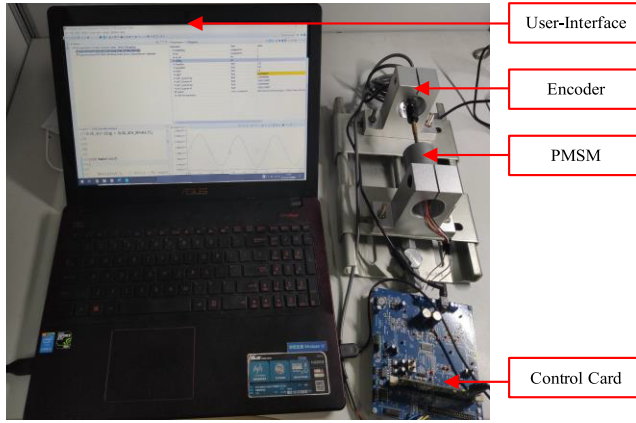
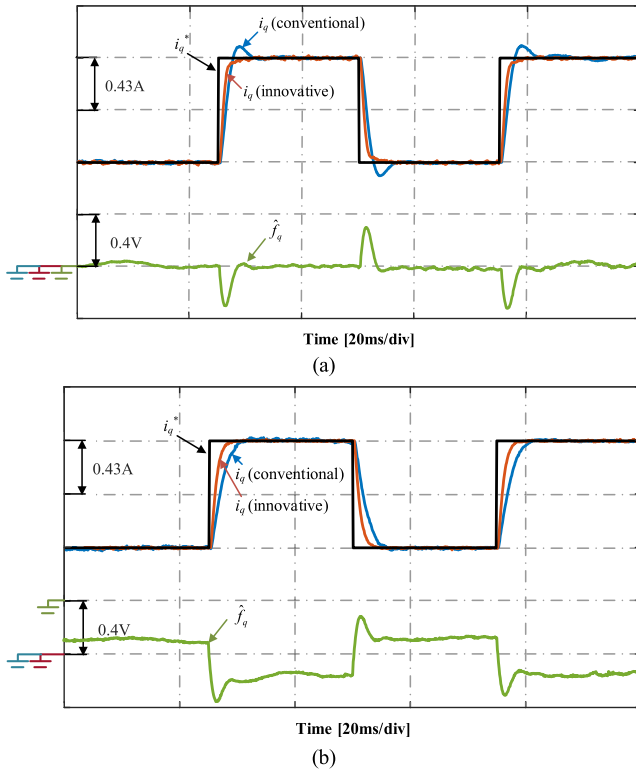


FIGURE 9. Experimental platform.

FIGURE 10. Tested square-wave current response of conventional IMC-based PI controller and the innovative current controller with parameters mismatch. (a) $L_s = 0.3L'_s$. (b) $R_s = 0.5R'_s$, $L_s = 0.3L'_s$.

A. ROBUSTNESS AGAINST PARAMETERS MISMATCH

A parameter mismatch appears by changing the machine parameters used in the proposed controller. Two set of experiments are conducted to verify the robust performance against the variations of parameters. In the experiments, $\alpha = 2500$ rad/s, $\omega_f = 4000$ rad/s are selected according to both tracking performance and sampling noises.

The machine runs in current-loop mode, and different types of current references are applied as input. Firstly, the square-wave (0.86A ~ 1.72A) current response of the conventional IMC-based PI controller and the innovative current

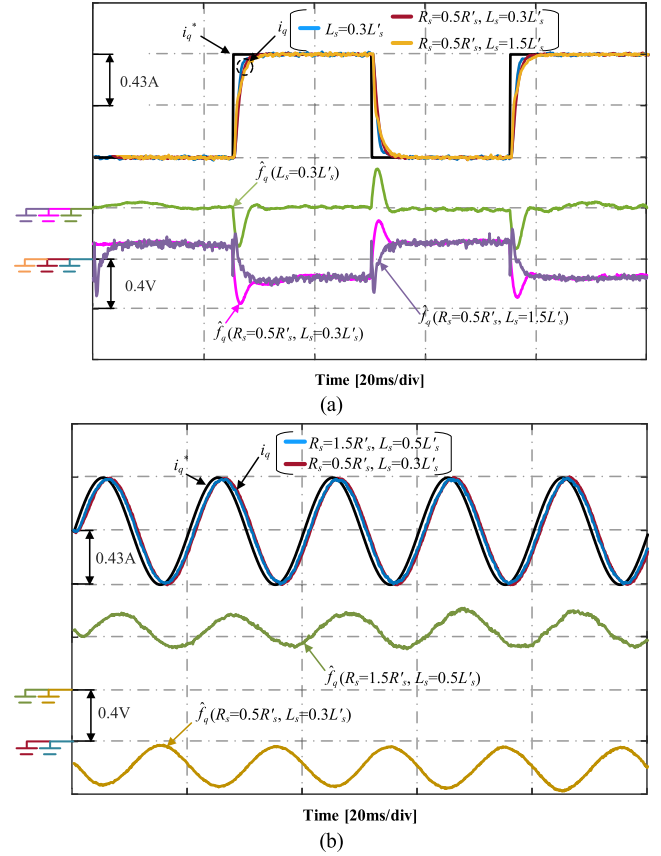
FIGURE 11. Tested results of the proposed current controller with different parameters mismatch. (a) Tested square-wave current response with $L_s = 0.3L'_s$ and $R_s = 0.5R'_s$, $L_s = 0.3L'_s$ and $R_s = 0.5R'_s$, $L_s = 1.5L'_s$. (b) Tested sinusoidal-wave current response with $R_s = 1.5R'_s$, $L_s = 0.5L'_s$ and $R_s = 0.5R'_s$, $L_s = 0.3L'_s$.

TABLE 4. Parameters of PMSM.

Parameters	Value
Rated voltage u_N	24 V
Rated speed n_N	4000 r/min
Pole pairs n_p	4
Stator resistance R_s	0.39 Ω
Stator inductance L_s	0.60 mH
Rotor flux linkage ψ_f	5.63e-3 Wb
Inertia J	1.69e-6 kg·m ²

controller are tested with two cases of parameters mismatch ($L_s = 0.3L'_s$ and $R_s = 0.5R'_s$, $L_s = 0.3L'_s$). The results in Fig. 10 demonstrate that the proposed method is less sensitive to the parameters mismatch due to the accurate disturbance estimation.

Secondly, the square-wave (0.86A ~ 1.72A) and the sinusoidal-wave (1.72+0.43sin(314t)A) current response of the proposed current controller with different parameters mismatch are tested and shown in Fig. 11.

In Fig. 11, the transient response is not affected by the parameters mismatch. Different mismatch disturbance can be estimated accurately and compensated satisfactorily, which

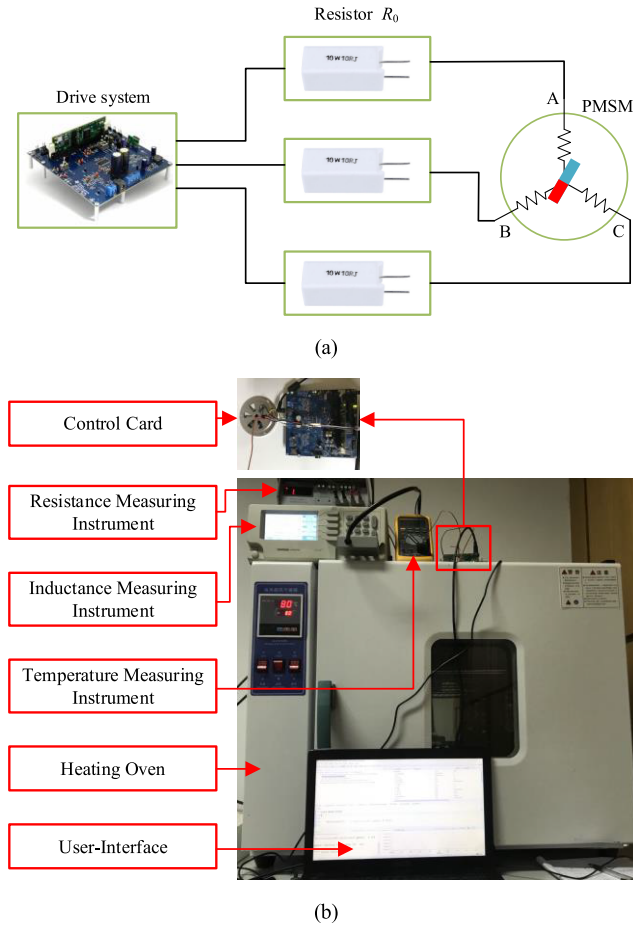


FIGURE 12. Experimental platform. (a) Method 1: Connect resistor in series with each phase winding of the machine. (b) Method 2: Place the machine into a heating oven.

guarantees the proposed method robustness to the parameters mismatch. Additionally, with the different reference current, the proposed method can effectively suppress the mismatch disturbance.

B. ROBUSTNESS AGAINST MACHINE PARAMETERS VARIATION

Fig. 12 explains two methods to change the machine parameters. Method 1: In Fig. 12(a), the resistance of stator winding increases by connecting a resistor ($R_0 = 0.5\Omega$) in series with each phase winding of the machine. Method 2: In Fig. 12(b), the motor is placed into a heating oven with a temperature range of 20°C to 80°C . The temperature measuring instrument is applied to detect the surface temperature of the motor. The resistance and inductance of motor at different temperature are measured separately.

In the first method, the winding resistance of the machine increases by 1.25 times. In the second method, the phase resistance and phase inductance at different temperatures are measured. The d - and q -axis components of the inductance are calculated by the method proposed in [22], and the results are shown in Fig. 13.

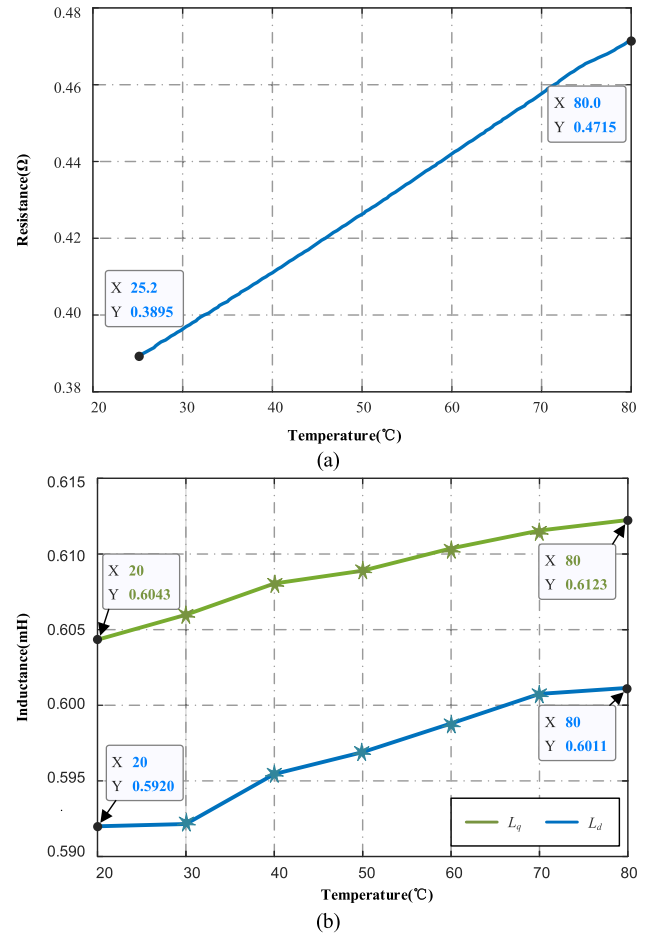


FIGURE 13. Motor parameters change with temperature. (a) Resistance. (b) Inductance.

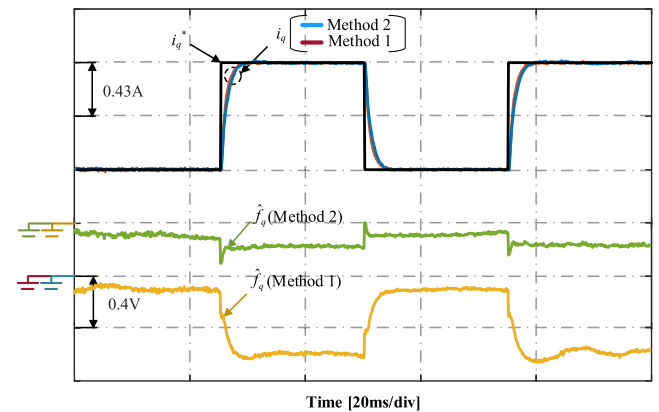


FIGURE 14. Tested results of the proposed current controller with different machine parameters variation.

The resistance increases by 21.1%, the d -axis inductance increases by 1.5%, and the q -axis inductance increases by 1.3% when the temperature changes from 20°C to 80°C . The square-wave (0.86A $\sim$ 1.72A) current response of the two cases are monitored and described in Fig. 14.

The proposed current controller can obtain an excellent current response in both cases without changing the controller parameters. The experimental results reflect that the proposed current controller has high robustness against parameters mismatch. Notably, decreasing the machine parameters used in the controller and increasing the actual machine parameters have the same effect on system by comparing the estimated disturbance in Fig. 14 with Fig. 11(a). Therefore, there is no doubt that the proposed current controller can work excellently in a complex operating environment.

VI. CONCLUSION

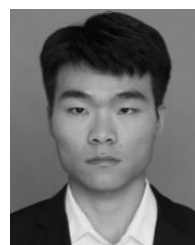
Since PMSM parameters vary with the environment temperature and operation condition, it is significant to design the current controller considering the time-varying properties of parameters. The conventional IMC-based PI current controller is not able to achieve satisfactory performance due to the parameter uncertainties from complex operating conditions. This paper utilizes an uncertainty and disturbance estimator to improve the control performance of the PMSM drive system. The proposed method has stronger rejection capability for parameters mismatch disturbance.

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